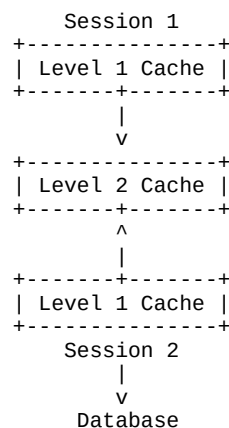


Chapter 7 - Hibernate Level 2 Cache & Query Cache

1. Introduction

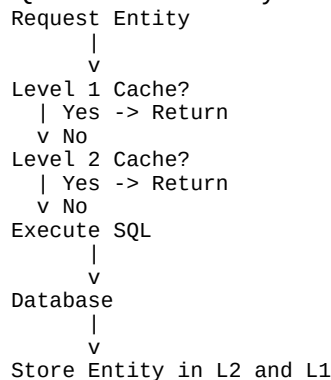
Unlike the Level 1 Cache, the Level 2 Cache is shared across Sessions. It is disabled by default and must be configured with a cache provider such as Ehcache, Infinispan or Caffeine (depending on your setup).

2. Level 2 Cache Architecture



3. How It Works

1. Session checks Level 1 Cache. 2. If not found, Hibernate checks Level 2 Cache. 3. If still not found, SQL executes. 4. Entity is loaded and stored in both caches (when configured).



4. Query Cache

Query Cache stores query result identifiers (typically primary keys), not complete entity objects. When Query Cache is enabled, Hibernate can reuse previous query results. The entities themselves are normally retrieved from the Level 2 Cache or the database if needed.

5. Cache Providers

Common cache providers include: • Ehcache • Infinispan • Caffeine (through supported integrations) Always choose a provider supported by your Hibernate version.

Feature	Level 1 Cache	Level 2 Cache
Enabled by Default	Yes	No
Scope	One Session	SessionFactory / Application
Shared Across Sessions	No	Yes
Stores	Managed entities	Entities (configured)
Destroyed	Session close	Application/SessionFactory shutdown or eviction

6. Concurrency Strategies

Common cache concurrency strategies are: • READ_ONLY • NONSTRICT_READ_WRITE • READ_WRITE • TRANSACTIONAL Choose the strategy based on how frequently the data changes.

7. Real-Time Scenario

Reference data such as Country, State, Currency or Product Category changes infrequently. Such entities are common candidates for the Level 2 Cache because many Sessions can reuse them.

8. Best Practices

- Use Level 2 Cache only for frequently read and infrequently updated data.
- Do not cache highly volatile data.
- Remember that Query Cache and Level 2 Cache serve different purposes.

9. Interview Questions

1. Difference between Level 1 and Level 2 Cache? 2. Is Level 2 Cache enabled by default? 3. What does Query Cache store? 4. Name common cache providers. 5. When should Level 2 Cache be avoided?